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CHAPTER III

EXPERIMENTAL SETUP

This chapter describes the experimental setup used to collect the data for the long-lived particle searches presented in Sections VI and VII. All results presented in this thesis are based on data collected by the ATLAS detector at the Large Hadron Collider (LHC) at CERN.

3.1 The Large Hadron Collider

The Large Hadron Collider (LHC) is currently the world's largest and highest-energy particle accelerator, with a total circumference of approximately 27 km. It is located about 100 m underground, straddling the border between Switzerland and France near Geneva. The LHC accelerates protons to nearly the speed of light using radio-frequency (RF) cavities and superconducting magnets, and brings counter-rotating proton beams into collision at four interaction points evenly spaced around the accelerator ring. At each interaction point resides one of the four main experiments: ATLAS, CMS, LHCb, and ALICE. The center-of-mass energy of proton–proton collisions is 13 TeV during Run 2 and 13.6 TeV during Run 3.

Protons are produced from hydrogen gas and accelerated through a sequence of injector machines prior to entering the LHC. Since hydrogen is initially in molecular form, electrons are first added to form H^- ions. These ions are then passed through a stripping foil, removing the electrons and producing protons (H^+). The protons are initially accelerated by RF cavities and focused using quadrupole magnets in the Linear Accelerator 4 (LINAC4), reaching an energy of 160 MeV.

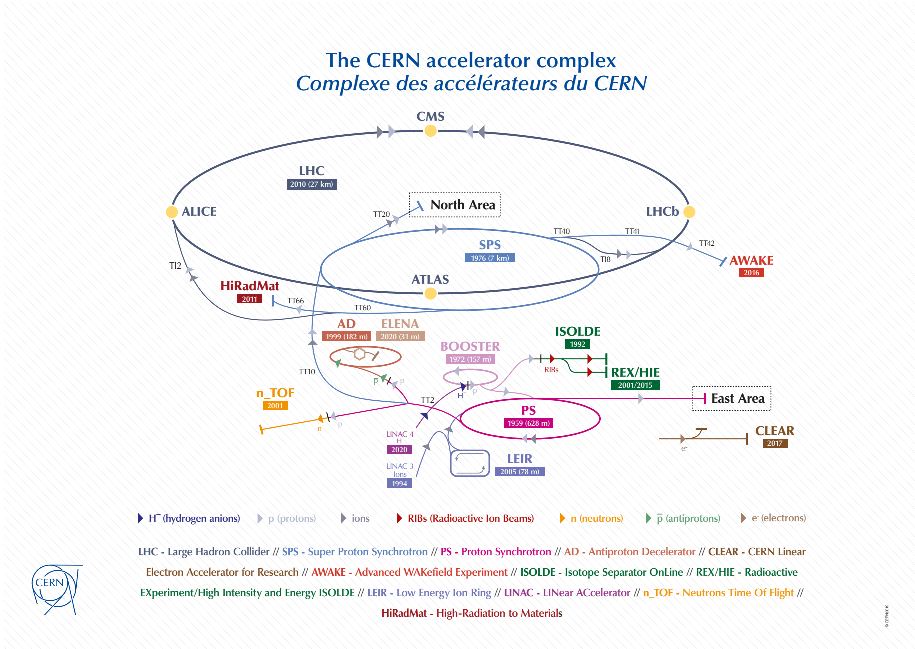

Figure 1. Schematic of the LHC accelerator complex**?**, showing the sequence of accelerators used to prepare and inject proton beams into the LHC ring. Image credit: CERN .

Subsequently, the proton beam is injected into the Proton Synchrotron Booster (PSB), where it is further accelerated to 2 GeV. From there, the beam is transferred through the Proton Synchrotron (PS) and the Super Proton Synchrotron (SPS), where it undergoes additional acceleration and bunch shaping before final injection into the LHC. A detailed description of the injector chain performance can be found in Ref. **?**.

Finally, the prepared proton bunches are injected into the LHC, where they are accelerated to their final energies and brought into collision at the four interaction points.

3.2 The ATLAS Detector

The ATLAS detector is a general purpose particle detector located at one of the four interaction points of the Large Hadron Collider (LHC). It is approximately

46 m long, 25 m in diameter, weighs about 7000 tons, and consists of multiple subsystems arranged concentrically around the interaction point.

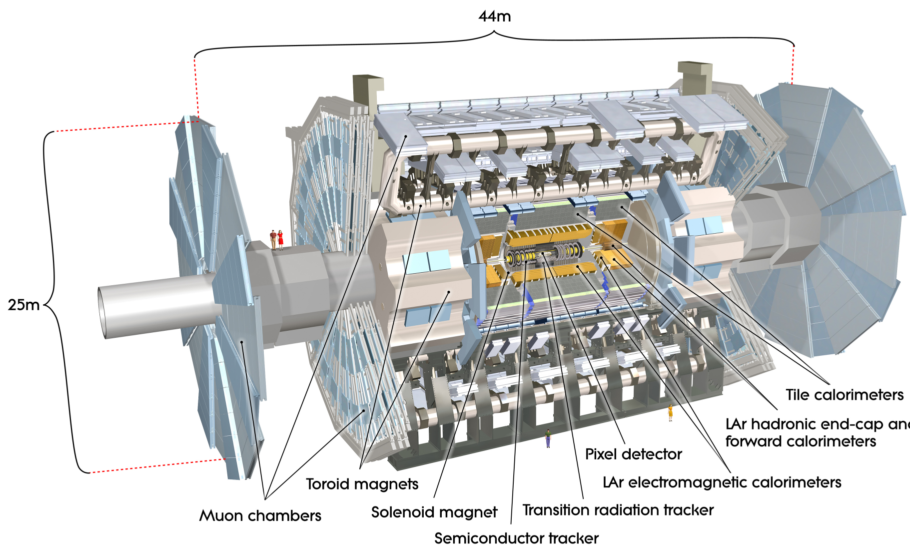

Figure 2. Schematic view of the ATLAS detector**?**, showing the main subsystems arranged around the interaction point. Image credit: CERN .

Starting from the interaction point and moving radially outward, the main subsystems are the Inner Detector (ID), the central solenoid magnet, the electromagnetic and hadronic calorimeters, the toroidal magnet system, and the Muon Spectrometer (MS). Each subsystem is designed to measure specific properties of particles produced in proton–proton collisions.

The Inner Detector is the closest subsystem to the interaction point and is responsible for reconstructing the trajectories of charged particles. It provides precise measurements of track positions and momenta within a 2 T magnetic field generated by the central solenoid, which surrounds the ID. Outside the solenoid lies the electromagnetic calorimeter, which measures the energies of electrons and photons through the development of electromagnetic showers. Surrounding this is the hadronic calorimeter, designed to measure the energies of hadrons via hadronic showering, working in conjunction with the electromagnetic calorimeter to provide a complete measurement of the event energy flow.

Enclosing the calorimeters is the toroidal magnet system, which supplies the magnetic field required for the Muon Spectrometer. The MS forms the outermost layer of ATLAS and is optimized for the detection and momentum measurement of muons. Due to their relatively large mass and weak interactions with matter, muons typically traverse the inner detector and calorimeters with minimal energy loss, making the MS essential for their identification. The MS employs a combination of tracking chambers and magnetic fields to measure the curvature of muon trajectories, enabling precise momentum determination.

The ATLAS detector uses a right handed coordinate system. The origin is at the nominal interaction point with the Z axis along the beam pipe. The Y axis points upwards and the X axis points to the center of the LHC ring. In ATLAS the normal angles θ and ϕ are used where θ is the polar angle measured from the positive z axis and ϕ is the azimuthal angle measured in the x-y plane from the x axis. One important variable used in ATLAS is pseudorapidity, η, which is defined as $\eta = -ln(tan(\theta/2))$ where θ is the polar angle measured from the positive z axis. Pseudorapidity is useful for ATLAS as it is invariant under Lorentz boosts along the beam axis with the assumption that the particles momentum is much larger then mass, $pc \gg mc^2$.

3.2.1 Inner Detector. The Inner Detector (ID) is the innermost subsystem of ATLAS and plays a central role in the long-lived particle searches presented in this thesis. It consists of three main components: the Pixel Detector, the Semiconductor Tracker (SCT), and the Transition Radiation Tracker (TRT).

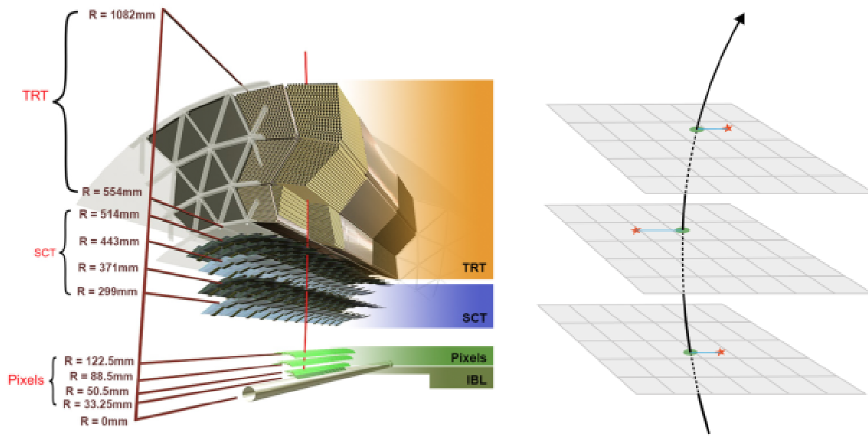

Figure 3. Schematic view of the Inner Detector**?**, showing the beam pipe, pixel barrel, SCT barrel, and TRT. Image credit: CERN .

3.2.1.1 Pixel Detector. The Pixel Detector is located closest to the interaction point and is composed of multiple layers of silicon pixel sensors. It includes the Insertable B-Layer (IBL), three barrel layers, and three endcap disks on each side of the detector. This configuration provides up to four space-point measurements for charged particles within a pseudorapidity coverage of $|\eta| < 2.5$.

The innermost pixel layer, known as the Insertable B-Layer (IBL), was installed during the first long shutdown of the LHC to enhance tracking and vertexing performance. The IBL is positioned at a radius of 33.5 mm from the beamline, enabling high-precision measurements close to the interaction point. It is constructed from 14 staves, each containing 32 front-end chips mounted on silicon sensors.

Two sensor technologies are employed in the IBL: planar sensors and 3D sensors. Planar sensors are used in the central region of each stave, while 3D sensors are utilized in the forward regions due to their superior radiation tolerance. Each sensor is segmented into pixels with dimensions of 50 μm in ϕ and

250 μm in z. Each stave contains 1280 columns and 336 rows of pixels, resulting in approximately 12.8 million pixels in the full IBL.

The remaining three barrel layers of the Pixel Detector, referred to as the B-layer, Layer 1, and Layer 2, are located at radii of 50.5 mm, 88.5 mm, and 122.5 mm, respectively. In these layers, planar silicon sensors with a thickness of 250 μm are used, with pixel dimensions of 50 μm in ϕ and 300 μm in z.

3.2.1.2 Semiconductor Tracker. The Semiconductor tracker(SCT) is the next subsystem after the Pixel Detector. The SCT is a precision microstrip detector that provides tracking at radial distances of 299 mm to 560 mm from the beam line at a $|\eta| < 2.5$. The SCT can provide momentum measurements of up to $\sigma pT/pT = 0.05\% \times pT[GeV] \oplus 1\%$

3.2.1.3 Transition Radiation Tracket.

3.2.2 Calorimeters. ATLAS employs two main calorimeter systems: the electromagnetic calorimeter and the hadronic calorimeter. Together, these systems are designed to absorb and measure the energies of particles emerging from the interaction point after they traverse the Inner Detector.

3.2.2.1 Electromagnetic Calorimeter. The electromagnetic calorimeter is a sampling calorimeter that uses lead as the absorber material and liquid argon as the active medium. Incident electrons and photons initiate electromagnetic showers in the lead absorbers, with the resulting ionization signals collected in the liquid argon layers to provide precise energy measurements.

The electromagnetic calorimeter is divided into a barrel section and two endcap sections. The barrel region covers pseudorapidities of $|\eta| < 1.475$, while the endcap regions extend the coverage to $1.375 < |\eta| < 3.2$.

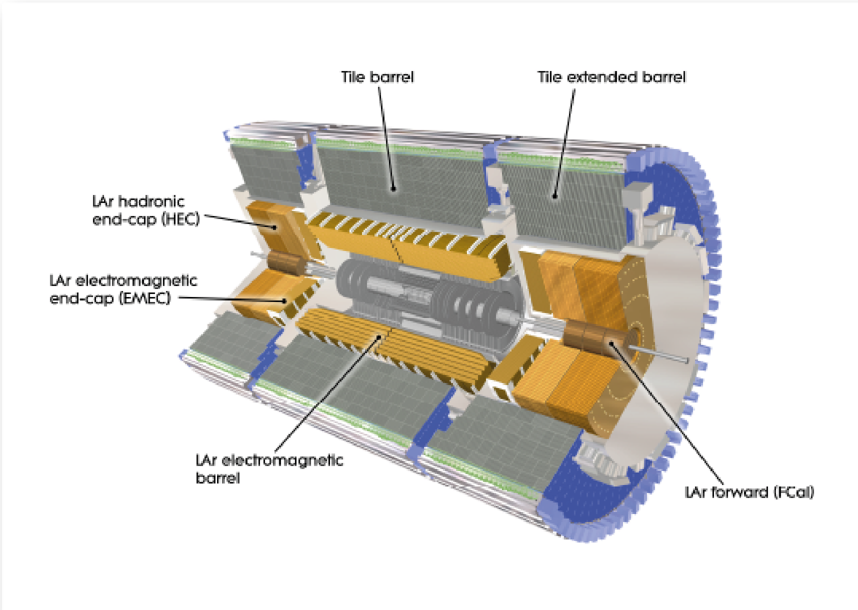

Figure 4. Schematic view of the Calorimeters**?**, showing the electromagnetic and hadronic calorimeters. Image credit: CERN .

3.2.2.2 Hadronic Calorimeter. The hadronic calorimeter is also a sampling calorimeter, employing steel and copper as absorber materials and scintillating tiles and liquid argon as active media. It is designed to measure the energies of hadrons through the development of hadronic showers.

Similar to the electromagnetic calorimeter, the hadronic calorimeter is divided into a barrel section and two endcap sections. The barrel region covers $|\eta| < 1.4$, while the endcap regions extend the coverage to $1.4 < |\eta| < 4.8$. In the forward and endcap regions, liquid argon technology is used to accommodate the higher radiation levels.

CHAPTER IV

EXPERIMENTAL UPGRADES

4.1 Phase 2 upgrades

4.1.1 ATLAS ITk.

CHAPTER V

OBJECT RECONSTRUCTION

CHAPTER VI

DISAPEARING TRACKS

CHAPTER VII

DISAPPEARING TRACKS WITH HIGH IONIZATION LOSS

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APPENDIX A

THE FIRST APPENDIX

A.1 Appendix One Section One

A.1.1 Chapter four section one sub-section one.

APPENDIX B

THE SECOND APPENDIX

B.1 Appendix Two Section One

B.1.1 Chapter two section one sub-section one.

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